

WOOSEOK KIM

Email: wooseok@dataplatforms.ca ◇ Phone: +1 613-292-5851 ◇ GitHub: 3seoksw
LinkedIn: <https://linkedin.com/in/wooseokkim-3seoksw> ◇ Ottawa, ON, Canada

EDUCATION

- Carleton University; Ottawa, Canada.** September 2025 – August 2027
M.S. Computer Science (Data Science Specialization) CGPA: 11.4 / 12
- San Jose State University; San Jose, California, United States.** July 2024 – August 2024
Certificate of Achievement: Silicon Valley Software Technology & Artificial Intelligence Program
- Sangmyung University; Seoul, Republic of Korea.** March 2019 – February 2025
B.S. Computer Science GPA: 97.0 / 100

EXPERIENCE

- Associate Data Engineer** May 2026 – Present
Revolution Data Platforms
- Associated to support the UHN migration from GCP to Azure.
- Research Assistant** September 2025 – Present
Intelligent Machines Lab (iML), Carleton University
- Advisor: Professor Majid Komeili
 - Exploring thesis directions in agentic AI systems.
- Undergraduate Research Intern** June 2023 – December 2024
Network Information Systems Lab, Sangmyung University
- Advisor: Professor KyungSeop Shin
 - First-author of a SCOPUS-indexed journal paper, titled "Optimal Power Allocation and Sub-optimal Channel Assignment for Downlink NOMA System Using Deep Reinforcement Learning".
 - Conducted comparative analysis of Bayesian uncertainty quantification methods on NASA's CMAPSS turbofan degradation dataset, evaluating prediction reliability under sensor noise.
 - Designed and implemented policy-based motion planning for an autonomous vehicle simulator using Deep Reinforcement Learning, and led technical seminars for lab members.

PROJECTS

- Ottawa Crime Rates and City Light Levels Analysis** January 2026 – April 2026
- Built a geospatial data pipeline using Pandas, cleaning and integrating crime reports with street-light datasets.
 - Applied statistical modelings (Difference-in-Differences, Pearson's Correlation Coefficient) to evaluate relationships between street-light infrastructure and crime trends.
 - Developed a crime prediction model (F1-score: 0.81) using an attention module with features selected through statistical analysis of crime patterns.
 - Performed model interpretability analysis using attention heatmap visualizations to uncover spatial and temporal patterns influencing crime predictions.
 - Developed visual analytics and dashboards using IBM Cognos Analytics and Matplotlib to support data-driven insights.
- Anomaly Detection in Jet Engine Dataset (CMAPSS)** January 2026 – April 2026
- Applied Variational Autoencoders (VAEs) to anomaly detection on NASA's CMAPSS jet engine degradation dataset.
 - Utilized a latent mapping function as an anomaly score from negative Evidence Lower Bound (ELBO), yielding AU-ROC of 0.94.
 - Achieved a F1 score of 0.708 for detecting gradual anomaly samples.

Reinforcement Learning Projects

- Applied attention module for an AlphaZero-style agent in the board game Gomoku, and outperformed the vanilla AlphaZero approach achieving higher ELO rating ($\Delta\text{ELO} \approx 500$).
- Developed a Super Mario Bros. agent using Double DQN with Python, PyTorch, and Gymnasium environment, achieving human-level performance of 30 seconds of playtime.
- Established an autonomous vehicle simulation, reduced 20% in lane deviation by applying transfer learning from obstacle detection to lane detection.

RESEARCH

Optimal Communication Resource Allocation Using DRL

January 2024 – March 2025

Sangmyung University

- Proposed a deep reinforcement learning framework integrating the REINFORCE algorithm for joint resource allocation (JRA) in downlink NOMA systems.
- Identified limitations of the existing JRA method, particularly their poor performance in channel assignment under dynamic conditions.
- Conducted extensive simulation experiments to evaluate the effects of the environment size, neural model architecture, and hyperparameters on convergence behaviour and performance metrics.
- Demonstrated a 39% improvement in resource allocation performance using an on-policy policy-gradient approach.

Brain Tumor Detection Using PK-YOLO

September 2025 – December 2025

Carleton University

- Improved PK-YOLO in brain tumor detection by outperforming state-of-the-art performance in axial plane data by 6.4% recall and 3.8% mAP@50.
- Applied a Mixture-of-Experts framework with plane-specific backbone models to mitigate performance degradation across MRI planes.

Dynamic Soft Prompting Using LION

October 2025 – December 2025

Carleton University

- Extended the LION MLLM by proposing a dynamic soft prompting mechanism to mitigate hallucinations in ViT tasks.
- Achieved VQA performance gains by 4.4% over LION, validating the effectiveness of dynamic prompting.
- Fine-tuned a BERT tag encoder using contrastive learning via parameter-efficient-fine-tuning (PEFT).

Battery Analysis Using Dynaformer

March 2024 – May 2024

Sangmyung University

- Implemented a Transformer-based Dynaformer model to predict lithium-ion battery voltage.
- Trained and evaluated on real-world datasets, achieving strong performance comparable to state-of-the-art performance.
- Analyzed the encoder representations, and captured battery aging and discharge characteristics via PCA.

HONOURS AND AWARDS

Research Assistant Scholarship	September 2025 – August 2027
Academic Scholarships for Computer Science (Sangmyung University)	2019, 2022, 2023, 2024
Outstanding Program Performance (San Jose State University)	August 2024
Entrepreneurship Award (San Jose State University)	August 2024
Outstanding Computer Science Graduation Project Exhibition Award (Sangmyung University)	November 2024
Fourth Prize in Capstone Design Competition (Sangmyung University)	December 2024
Software Talent Scholarship (Sangmyung University; SW-Centered University Project Group)	December 2024
Army Warrior Award (Republic of Korea Army)	October 2021
The Army Commendation Medal: Outstanding Service (U.S. Army)	August 2021
Outstanding KATUSA Award (Republic of Korea Army)	May 2021
Excellence Award in Combined Command Post Training (U.S. Army)	September 2020

TECHNICAL SKILLS

- Programming Languages: Python, C, Java, C++, Rust, JavaScript, TypeScript, React
- Machine Learning Frameworks: PyTorch, Tensorflow, Keras, PyTorch Lightning
- Data Science: Pandas, NumPy, GeoPandas, SQL
- Tools: IBM Cognos Analytics, Linux, Git, Docker, Anaconda, Jupyter Notebook, $\LaTeX$

OTHERS

Military Service

April 2020 – October 2021

Korean Augmentation to the United States Army (KATUSA)

- Korea Army Squad Leader (Republic of Korea Army)